

Addressing the Hydrogen Transfer Limitation in High Concentration Nitrate Reduction by Ruthenium Promoter

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Abstract: The electrochemical synthesis of ammonia via nitrate reduction reaction (NO₃RR) has been intensively researched as an alternative to traditional Haber-Bosch process. However, industrial viability is challenged by low concentration NO₃RR owing to the mismatch between hydrogenation and nitrate reduction in a nitrate-rich condition. Herein, we demonstrate that by

engineering ruthenium clusters on the surface of Cu_2O catalyst, the rate of NO_3RR can be enhanced significantly to achieve a 10.0 A operation in 1 M nitrate electrolyte, with a 100% Faradaic efficiency toward ammonia (6.87 USD per 1 kg of ammonia) in a 16 cm^2 flow electrolyser. Detailed mechanistic studies by deuterium labelling, operando Fourier-transform infrared spectroscopy, and theoretical simulations reveal enhanced hydrogen adsorption and transfer from ruthenium clusters to the Cu_2O matrix for the hydrogenation of nitrate adsorbents, and high faradaic efficiencies can be maintained up to 3 M nitrate. Our work sheds lights on the importance of engineering synergistic interactions between co-catalysts for addressing the kinetic bottleneck in electrosynthesis.

Introduction

Electrolysis in a concentrated electrolyte (≥ 2 M) is required for economically viable industrial production in view of unusual mass transfer and reaction kinetics, wider electrochemical window, and less generation of waste solutions¹⁻⁵. Successful deployment of concentrated electrolytes (*e.g.*, 1 to 6 M KOH aqueous solution) have been demonstrated in alkaline fuel cells, seawater electrolyzers and zinc-air batteries⁶⁻⁸. However, the switching of dilute to concentrated electrolyte does not follow simple linear correlation between performance and electrolyte concentration. Unlike the presence of solvent-separated ion pairs and free solvent molecules in dilute electrolyte, contact ion pairs and cation-anion aggregates are usually involved in extremely concentrated electrolyte due to the scarcity of solvent molecules^{9,10}. Such unique solvation chemistry leads to significantly different electrochemical profiles and reaction kinetics. Particularly, the availability and mobility of proton in aqueous electrolysis are dependent on salt-induced perturbation of hydrogen bonding network and formation of strong water-ion pairs, influencing the hydrogen acquisition and transfer kinetics^{11,12}.

Industrial adoption of electrochemical nitrate reduction reaction (NO_3RR) as an alternative to the Haber-Bosch process requires quantitative conversion of concentrated nitrate (up to 2 M in industrial and agricultural wastewater) for ammonia electrosynthesis¹³⁻¹⁶. The difficulty in operating with high concentration of nitrates is a bottleneck preventing the process to be adopted¹⁷⁻²¹. The conundrum is that good NO_3RR catalysts (such as Cu_2O and other Cu-based

catalysts) typically have strong binding interactions with nitrates^{19,22,23}, which inevitably results in nitrate poisoning of the adsorption site for atomic hydrogen generated from hydrogen evolution reaction (HER), preventing subsequent hydrogenation steps in NO₃RR. The mismatched reaction kinetics in HER and NO₃RR has hampered the use of concentrated electrolyte in NO₃RR, urging for the design of co-catalytic platform to optimize the reaction kinetics^{24,25}. For instance, the introduction of secondary metals (such as Ni, Pd, Rh and Ru)^{18,26-29} into copper-based catalyst provides extra hydrogen binding sites and reduces the reaction barriers in hydrogenation steps. The catalytic mechanism could be sensitive to ion concentration and local electric field, for example in dilute electrolyte, it has been reported that Ru single atoms, Ni or Ru dopants serve as the catalytic active sites for nitrate.^{18,26,27} The underlying mechanism with respect to the concentration dependence remains unclear.

Herein, we have chosen a benchmark NO₃RR catalyst (ruthenium modified cuprous oxide, Ru-Cu₂O) to illustrate the working principle of reaction kinetics regulation by secondary metals. The correlation between electrolyte concentration and hydrogen transfer kinetics is carefully studied in the H-cell setup from very diluted (0.01 M) to nearly saturated electrolytes (3 M). Ruthenium clusters served as hot spots in hydrogen adsorption from bulk electrolyte and promotes subsequent hydrogen transfer to the Cu₂O matrix for the hydrogenation of nitrate adsorbents, thus enabling much enhanced activities at high concentrations by pairing reaction kinetics. When assembled in a 16 cm² flow electrolyzer, our Ru-Cu₂O catalyst allows an ammonia production cost of 6.87 USD per kilogram of ammonia, yielding a remarkable ~100% Faradaic efficiency toward ammonia in 1 M nitrate at 10.0 A full cell current.

Results and discussion

Geometry and electronic configuration of benchmark Ru-Cu₂O catalyst. As depicted in **Figure 1**, we have synthesized ruthenium cluster-decorated Cu₂O nanowires (Ru-Cu₂O) catalyst for NO₃RR at high nitrate concentrations, where ruthenium domains serve as reaction hot-spots to promote hydrogen acquisition and transfer kinetics. Typical preparation of Ru-Cu₂O involves anodic electrochemical deposition of Cu(OH)₂ nanowires on Cu foam, impregnation with RuCl₃ precursor and subsequent annealing in a hydrogen/argon mixture to furnish the Ru-Cu₂O³⁰. The metal

loading of Ru on Cu₂O was determined as 0.91 wt% from inductively coupled plasma optical emission spectroscopy (ICP-OES). Uniform distribution of Cu₂O nanowires on Cu foam is observed in the scanning electron microscopy (SEM) images in Figure S1, showing a (111) facet in high resolution transmission electron microscopy (HRTEM), and the predominance of (111) facets is also reflected in X-ray diffraction (XRD) in Figure S2 and S3. EDS mapping in Figure S6 confirms the existence of Ru throughout the catalyst matrix. The surface coverage of Ru on the Cu₂O is calculated as ~ 5% by comparing the electrochemical active surface area (ECSA) from hydrogen underpotential deposition and capacity testing in Figure S7 and S8. For comparative study of the role of Ru and Cu₂O in NO₃RR, we also prepared a reverse structure where Cu₂O adopts the form of clusters on a Ru substrate, where the exposed coverage of Ru is higher than that of Cu₂O with a thick, fully-covered Ru layer by overpotential electrodeposition in the RuCl₃ solution (Figure S1).

The valence state of Ru-Cu₂O was examined by X-ray absorption near-edge structure (XANES) and X-ray photoelectron spectroscopy (XPS). The Ru *K*-edge XANES spectrum of Ru-Cu₂O reveals a slightly positive charge (higher white line intensity than Ru foil) in Figure 2c owing to charge transfer between Ru and Cu₂O. Likewise, spin-orbit doublets attributed to positively charged Ru species (Ru^{σ+}) could be observed at 462.08 and 484.86 eV in the XPS Ru_{3p} spectrum in Figure 2e³¹. This is reflected in the peak shifting of 0.3 eV in the XPS Cu_{2p} spectra after Ru decoration (Figure 2f). Further analysis on the coordination environment of Ru-Cu₂O by extended X-ray absorption fine structure (EXAFS) spectroscopy confirms the clustering of Ru in our samples. A prominent Ru-Ru peak at 2.4 Å was observed in the Fourier transform-EXAFS spectra in Figure 2d.

Co-catalytic role of ruthenium clusters. It is well-known that apparent activity of catalyst usually does not scale linearly with the concentration of electrolyte, particularly for those reactions involving multiple proton-coupled electron transfers³². The kinetic limitation arises owing to mismatched capability in the activation of two or more reactants (such as hydrogen and nitrate in NO₃RR).

The importance in regulating the reaction kinetics is reflected in the yield of ammonia using various catalysts at nitrate concentration ranging from 0.01 M to 3 M in Figure 3a. The production rate of ammonia using Cu₂O catalyst increases with concentration in the dilute regime (0.01 to

0.1 M) due to promoted mass transfer (Fick's law) and reaction kinetics. However, the apparent activity of Cu_2O experiences a decline of 32% from 0.1 to 3 M owing to poisoning of surface sites by nitrates. Similar phenomenon has been observed in the Pt-catalyzed NO_3RR ³³, where a negative reaction order for the rate-determining step (RDS) in the conversion of nitrate to nitrite was found at high concentrations (> 0.1 M). This indicates the quasi-equilibrium of nitrate adsorption and conversion on the catalyst surface. Reactive species such as atomic hydrogen ($^*\text{H}$) must compete for surface sites under nitrate-rich condition. Therefore, our hypothesis is whether by introducing ruthenium hot-spots on Cu_2O surface, we can provide a sink for atomic hydrogen³⁴. As shown in **Figure 3a**, our Ru- Cu_2O enables a remarkable 13-fold increase in ammonia production rate when increasing the nitrate concentration from 0.01 to 1 M. No significant performance decay is observed until a very high concentration of 3 M. It is worth mentioning that 3 M is close to the saturated solubility of KNO_3 at 298.15 K (3.76 M). Control samples using fully covered Ru CL- Cu_2O (Figure S16) and reversed Cu_2O -Ru CL- Cu_2O (Figure S17) exhibit no major difference with pure Cu_2O catalyst, suggesting that a particular spatial segregation profile of Ru and Cu_2O is necessary to optimize the rate of nitrate and hydrogen activation.

Detailed investigation on the NO_3RR performance of Ru- Cu_2O was conducted in a H-cell setup at various concentrations. The production rate of ammonia was quantified by colorimetric and NMR methods in Figure S10. As shown in **Figure 3b**, Ru- Cu_2O achieves a nearly 100% Faradaic efficiency toward ammonium at -0.1 V and -0.2 V in 0.1 M nitrate, and maintains a $> 90\%$ Faradaic efficiency at -0.4 V at a current density of 1.26 A cm^{-2} . At higher concentration (1 M), Ru- Cu_2O exhibits an almost 1.5 times higher current density (2.01 vs. 1.39 A cm^{-2}) and Faradaic efficiency (75% vs. 49%) than pure Cu_2O between 0 V and -0.4 V. This corresponds to a remarkable ammonium yield rate of $7.02 \text{ mmol h}^{-1} \text{ cm}^{-2}$ in **Figure 3c**. ^1H NMR spectra were also collected for Ru- Cu_2O catalyst at -0.2 V to confirm the origin of ammonia using by ^{15}N isotope labeling. As shown in Figure S11, the electrolyte after K^{15}NO_3 reduction reveals a typical doublet of $^{15}\text{NH}_4^+$ at 6.86 and 7.00 ppm, without the presence of the triplet of $^{14}\text{NH}_4^+$ in the ^1H NMR spectra. This confirms the formation of ammonium originated from nitrate rather than nitrogen in air³⁵. The concept of regulating the hydrogen and nitrate activation can be further extended to other secondary metals (such as Pd). The Pd- Cu_2O catalyst exhibits a similar enhancement to Ru- Cu_2O owing to strong hydrogen

adsorption properties, giving a maximum ammonium yield rate of $4.71 \text{ mmol h}^{-1} \text{ cm}^{-2}$ in 1 M nitrate (Figure S18). However, further increasing the metal loading of Ru to 3.6 wt% (Figure S19) will lead to over-competition of HER against NO₃RR, resulting in a 40% decline in the current density in 0.1 M nitrate (Figure S20).

To validate the influence of hydrogen adsorption and transfer kinetics on nitrate reduction, we have performed HER and NO₃RR measurements in water and deuterium oxide using linear sweep voltammetry (LSV). Kinetic isotope effect (KIE) values are expressed by the ratio of current density measured in H₂O and D₂O. Typically, a KIE value greater than 1 implies the proton-coupled transfer step (primary KIE), rather than electron transfer step, is rate limiting during the electrochemical reactions (secondary KIE). As shown in Figure S12, the incorporation of Ru leads to a higher energy barrier (higher KIEs) for hydrogen evolution compared to pure Cu₂O.³¹ In the presence of nitrate (**Figure 3d**), pure Cu₂O exhibits a much higher KIE value of 3.08 at -0.4 V (vs. ~ 2 in HER in the absence of nitrate). This implies the difficulty in the acquisition of protons or atomic hydrogen from water splitting. In contrast, Ru-Cu₂O retains a considerably low KIE value of 1.7 under the same condition. We speculate that this may be due to the hydrogen trapping effect of Ru hotspots. The lower KIE value in NO₃RR compared to HER for Ru-Cu₂O also suggests favorable hydrogen transfer to nitrate rather than the recombination in hydrogen evolution. The KIE values of Ru-Cu₂O in NO₃RR maintain a steady value throughout the measurement range, whose preference on NO₃RR explains the excellent Faradaic efficiency to ammonia compared to control catalysts. Further details on the electrochemical profile of various catalysts are provided in Figure S13 to S21.

We also employed *operando* Fourier-transform infrared spectroscopy (FTIR) to monitor the adsorption configuration of reactants and intermediates at different concentrations of nitrate. The setup for *operando* FTIR is shown in Figure S22. When operating in 0.1 M nitrate, only one peak assignable to the *NO species could be observed at $\sim 1612 \text{ cm}^{-1}$ for pure Cu₂O (Figure S23a), whereas an additional signal from the *NH₂OH species is seen at 1184 cm^{-1} for Ru-Cu₂O in Figure S23b^{33,36,37}. The appearance of hydrogenated intermediates on Ru-Cu₂O validates the promotion effect of Ru on the hydrogenation kinetics. At a lower concentration of 0.01 M, signals related to the *NO₂, *NHO and *NH₃ are detected in the FTIR spectra of Ru-Cu₂O owing to the abundance

of surface hydrogen (**Figure 3f**)³⁸. On the other hand, control sample using Cu₂O decorated Ru reveals a significantly decreased intensity for *NO and *H₂O (Figure S24). This implies that HER dominates over NO₃RR on the Ru-rich surface, which also reduces the faradaic efficiency for ammonia production. In conclusion, Ru clusters are highly effective in splitting water and capturing hydrogen while a delicate balance is necessary for pairing the HER and NO₃RR kinetics (**Figure 1**).

Mechanistic insights on the hydrogen acquisition and transfer. *Ab-initio* molecular dynamics (AIMD) simulation was performed at room temperature (300 K) to study how hydrogen is transferred to the *NO₃ intermediates. The major argument lies in the function of Ru in reaction pathway: a true active site for nitrate as claimed in previous works¹⁸, or a promoter on hydrogen transfer kinetics. As the calculation results in **Figure XX ~ XX** show, Cu₂O is a more favorable binding site for nitrate against Ru. This is supported by the poisoning effect at high nitrate concentrations and the negative influence in NO₃RR on a Ru-rich surface. The visualized MD pathway in **Figure 4a** suggests that water or hydroxide molecules play an important role in H transfer. Proton couples strongly with nearby H₂O and hydroxide molecules. Such effect increases the effective interacting area between hydrogen atoms and *NO₃ intermediates in Helmholtz layer. When the *NO₃ intermediate on Cu₂O captures two H from H-bonding networks (i to iii), it will generate H₂O and the *NO₂ intermediate (iii to iv). The hydrogenation of *NO₂ to *HNO₂ is believed as the rate determining step, which is difficult to inspect in the following MD process. Meanwhile, the formed hydroxide due to nitrate reduction to nitrite captures another H from Ru clusters to regenerate the H₂O for further hydrogen transfer (v to vi). Considering the abundance of water and hydroxide molecules in the electrolyte, the above processes occur extensively on the catalyst surface. Thus, hydrogen atoms can be readily transferred from Ru to *NO₃ in Helmholtz layer, which is responsible for the promoted reaction rates.

Balancing the kinetics of HER and NO₃RR is important because excessive HER decreases the faradaic efficiency for ammonia production, yet if HER is suppressed too much, the rate of hydrogenation of nitrates will be negatively affected. At high concentration of nitrate, the surface sites for hydrogen adsorption become blocked, resulting in the suppression of surface-mediated hydrogenation process. To gain insight into how a co-catalytic system can address this problem,

we calculate the adsorption energies of hydrogen on the Cu₂O and Ru-Cu₂O in the presence of nitrates. We calculated the adsorption energy barrier of an additional atomic hydrogen on Cu₂O and Ru-Cu₂O with various nitrate coverages ranging from 1/12 to 1/4 monolayer (**Figure 4b**). For pure Cu₂O, there is no obvious variation in energy to adsorb another atomic hydrogen at a nitrate coverage of 1/12 or 1/6. However, a much higher adsorption energy (0.93 eV) appears when the nitrate coverage increases to 1/4 monolayer owing to their competitive adsorption. In contrast, the introduction of ruthenium hot-spots will reverse the trend in the adsorption energy. The H adsorption energy is -0.37 eV at 1/12 coverage, which decreases slightly to -0.25 eV at 1/4 coverage. This phenomenon further proves that ruthenium serves as reservoirs of surface hydrogen and promotes hydrogen transfer kinetics on the Cu₂O surface, which facilitates high concentration nitrate reduction.

Finally, we have conducted DFT calculations to identify the reaction pathway of NO₃RR on the Cu₂O and Ru-Cu₂O catalysts in **Figure 4c**. Both catalysts share the same reaction pathway from *NO₃ to *NO₂, *NHO, *NH₂OH and finally *NH₃ as demonstrated by *operando* FTIR analysis. The formation of *HNO₃ and *HNO₂ intermediates are found to be rate-determining steps in NO₃RR. Owing to the electron transfer from Ru, the free energy barrier is reduced by 0.07 and 0.14 eV in both rate-determining steps.

Industrial-level nitrate reduction in a 16 cm² flow electrolyzer. By optimizing the spatial distribution of Ru clusters on Cu₂O to achieve the best performance, we demonstrated the operation of flow electrolyzer at an industrial current level of 10.0 A with an active electrode area of 16 cm²³⁹. The flow rate of electrolyte was set to 3 mL min⁻¹. Monolithic Ru-Cu₂O catalyst (16 cm²) and dimensionally stable anode (DSA) were used as cathode and anode. As shown in **Figure 5a**, full cell current gradually increases with increasing cell voltages and nitrate concentrations, affording a maximum current of 10.0 A at 4.2 V in 1 M KNO₃. Further analysis on the Faradaic efficiency in **Figure 5b** reveals a nearly 100% Faradaic efficiency toward ammonia in such condition, corresponding to an ammonia yield rate of 46.7 mmol h⁻¹ and a single pass conversion of 26% from nitrate into ammonia. At lower concentrations, the flow electrolyzer retains excellent Faradaic efficiency (73% at 0.32 A and 99% at 3.10 A for 0.01 and 0.1 M nitrate). The single pass conversion of flow electrolyzer greatly improves to 74% and 94% for the operation in

0.01 M and 0.1 M nitrate (**Figure 5e**). This outperforms the existing flow electrolyzer performance for NO₃RR in literature reports (Table S2).

Process durability is another key criterion for practical ammonia electrosynthesis. The stability of Ru-Cu₂O catalyst was examined in 1 M KNO₃ at a full cell voltage of 3.8 V in **Figure 5f**. Dynamic flow rate testing from 1 to 5 mL min⁻¹ was performed to exclude the influence of mass transfer limitation in our flow electrolyzer. After initial stabilization, a steady full cell current of ~7.50 A was obtained throughout the whole measurement (800 mins).

Finally, we have also estimated the electricity consumption and production cost of our ammonia production based on the market price of electricity in Singapore (0.12 USD per kWh). As shown in **Figure 5c**, an electricity consumption of 164.5 kWh or 21.4 USD in equivalent is needed to produce 1 kg of ammonia using 0.01 M nitrate. The operation at higher concentration (1 M) significantly reduces the production cost to 6.87 USD (50 kWh) per kilogram of NH₃ and affords a much wider operational window from 2.8 to 4.2 V. This is beneficial for actual operation in high-concentration nitrate wastewater.

Conclusion

In summary, we have investigated NO₃RR at high molar concentration of nitrate. By introducing ruthenium hot-spots on Cu₂O catalyst to form a co-catalytic system, the reaction kinetics of NO₃RR and HER can be paired well, which results in a substantially improved electrocatalytic performance. We demonstrated a NO₃RR flow electrolyzer that operates at 10.0 A with a nearly 100% Faradaic efficiency toward ammonia and a single pass conversion of 26% when using 1 M nitrate as electrolyte. Our study highlights the importance of pairing the reaction kinetics of NO₃RR and HER in a co-catalytic system for productive electrosynthesis.

Experimental Section

Electrochemical deposition of Cu(OH)₂ nanowires on Cu foam: Cu foam was rinsed by 1 M HCl solution and DI water. A two-electrode system was used, where the copper foam was used as the anode and platinum wire electrode was used as the cathode, and a current of 10 mA cm⁻² was applied for 20 mins without stirring in an electrolyte of 100 mL of 3 M NaOH solution. After

deposition, the entire copper foam changed into blue color. The $\text{Cu}(\text{OH})_2$ nanowires deposited Cu foam was washed by DI water and dried at room temperature.

Ru decorated Cu_2O catalyst on Cu foam (Ru- Cu_2O): The $\text{Cu}(\text{OH})_2$ -grown Cu foam was immersed in 100 mL of 0.25 mM RuCl_3 for 2 hours at 500 rpm stirring, and washed by DI water for three times before drying at room temperature. The sample was annealed in a gas mixture of hydrogen and argon (50:50 sccm) at 200 °C for 2 hours at a heating rate of 10 °C min^{-1} to obtain the Ru- Cu_2O catalyst. Unmodified Cu_2O and other metal-modified (such as Pd) Cu_2O catalysts were prepared by a similar method.

Electrochemical nitrate reduction in a flow-cell setup: Measurements were conducted in a 16 cm^2 flow electrolyser separated by a anion exchange membrane (SELEMION AMVN) at room temperature using an Arbin BT-G workstation, which has a maximum support current of 10 A [Caution: This exceeds the safe current limit for human and may cause **fatality** if not handling properly]. The flow cell was assembled with two stainless-steel cover plates, two gold-coated copper plates as current collector, two monopolar graphite plates with parallel paths for electrolyte distribution. Flow rate was controlled by two Watson Marlow 120S peristaltic pumps, and the actual flow rate was determined by measuring cylinder. A dimensionally stable anode (DSA, RuIr/TiO_2 on Ti-mesh, $4 \times 4 \text{ cm}^2$) and a catalyst modified Cu foam electrode ($4 \times 4 \text{ cm}^2$) were used for a two-electrode system in an aqueous electrolyte containing 1 M KOH and 0.01/0.1/1 M KNO_3 . Chronoamperometry (CA) was performed at constant potential. The flow-out solution from the catholyte chamber was immersed into a glass vial containing 6 mL of 1 M HCl to acidify the ammonia solution. Stability test was conducted at various flow rates for a continuous operation of 800 minutes at a full cell voltage of 3.8 V. The amount of ammonia from catholyte were also determined by the indophenol blue and NMR method.

Material characterization. The following equipment were used: STEM-HAADF (200 kV by FEI Titan Themis 60-300), EDS (200 kV by Super-X EDS system), TEM (200 kV by FEI Titan), SEM (JEOL JSM-6701F), XPS (Thermo Fisher Scientific K-Alpha⁺, monochromatic Al K_α), XRD (Bruker D8), NMR (Bruker Advanced 500 MHz), UV-Vis (Shimadzu UV-3600), ICP-OES (Perkin Elmer Avio 500, ppm level accuracy), Electrochemistry (Autolab PGSTAT30 & Arbin BT-G), *in-situ* FTIR (Bruker Vertex

80). Ru *K*-edge XAS: 250 mg of 0.91 wt% Ru-Cu₂O powder (without Cu foam) was finely ground using a mortar and pestle before being pressed into a 10 mm pellet. Measurements were carried out in the 7-BM beamline at the National Synchrotron Light Source II at Brookhaven National Laboratory in the fluorescence mode (Ru *K*-edge). Data analysis and simulation were carried out on Athena, Artemis, and Hephaestus (Version 0.9.23).

Data availability. All data are available from the authors upon reasonable request.

Supporting Information

Supporting Information is available online.

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AUTHOR CONTRIBUTIONS

Q.H. and O.P. conceived the research, synthesized the materials, conducted catalytic measurements under the supervision of Z.C. and K.P.L. *Operando* FTIR and XPS measurements were performed by M.L. and Z.X.; ab-initio molecular dynamics (AIMD) simulation was performed by K.Y. under the supervision of M.Y.; DFT calculation was performed by Q.H.; XAS measurements and data processing were conducted by Y.D., L.M.; Large current flow cell operation was performed with assistance from S. H. and Q. W; The draft was written by Q.H. and Z.C., and revised by K.P.L. All authors discussed and commented on the manuscript.

Competing financial interests: The authors declare no competing interests.

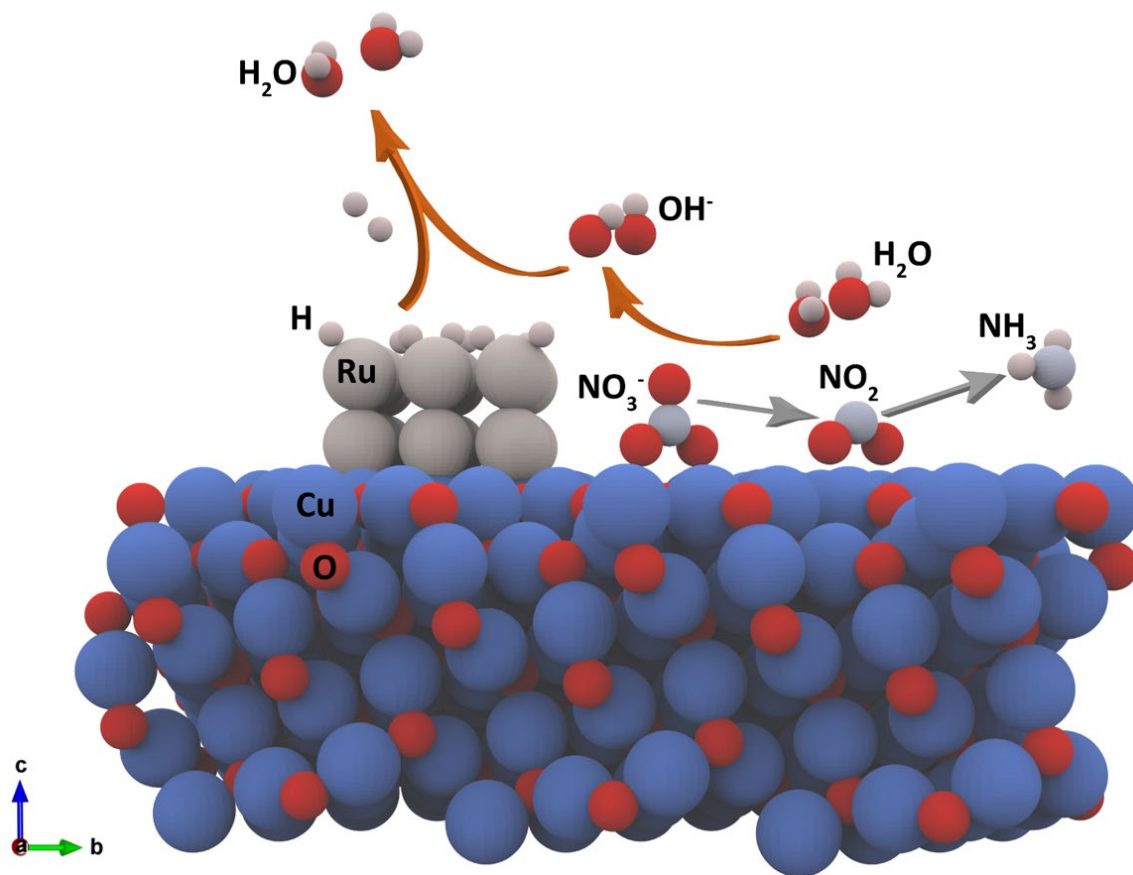


Figure 1. Schematic illustration. Ruthenium-mediated hydrogen acquisition and transfer from water molecules for efficient nitrate reduction. Color scheme: Cu atom in blue, O atom in red, Ru atom in cream, H atom in white, N atom in light purple. Direction of Cu_2O crystal (111 facet) is provided in the schematic.

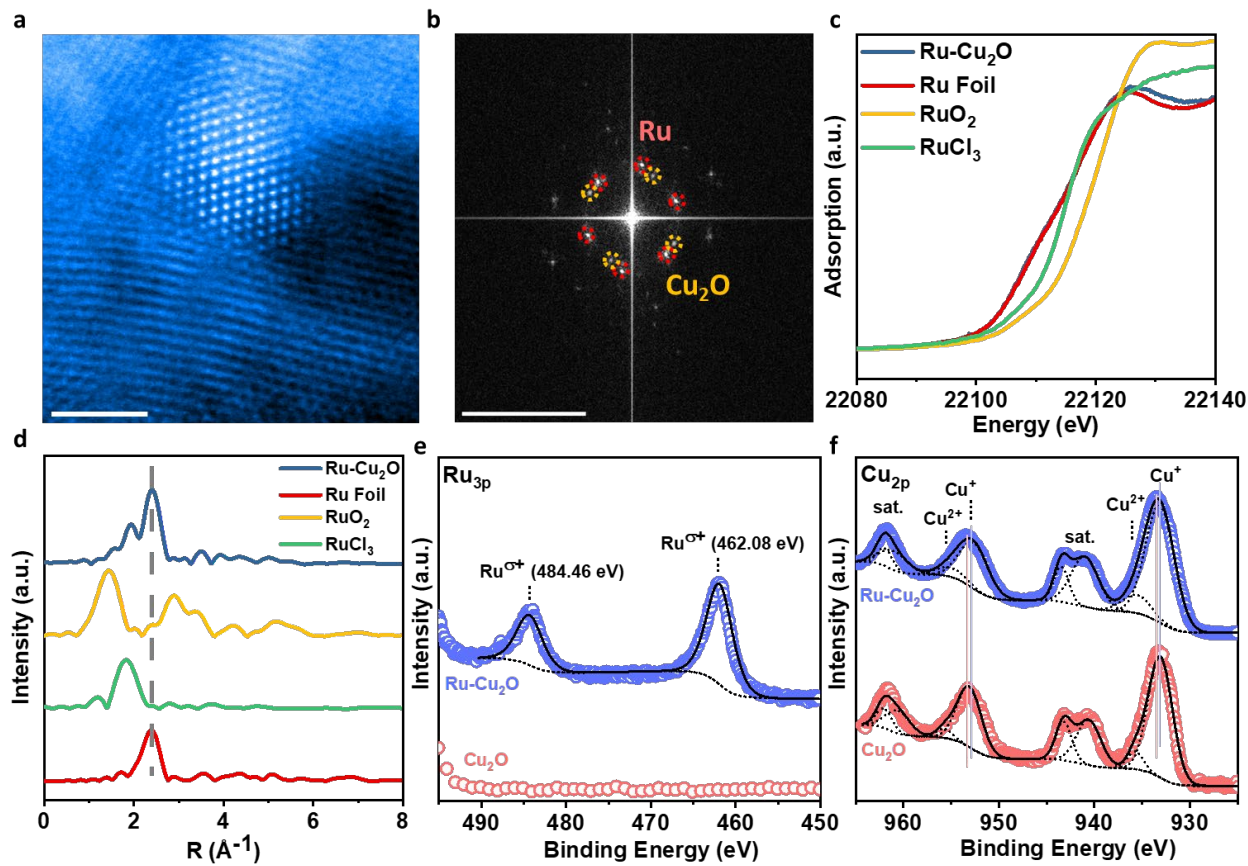


Figure 2. Characterization of the Ru-Cu₂O catalyst. (a) STEM-HAADF image of Ru-Cu₂O and (b) the corresponding FFT pattern showing the co-existence of Ru and Cu₂O; (c) Ru K-edge XANES and (d) EXAFS spectra of various catalysts; (e) XPS Ru_{3p} and (f) Cu_{2p} core-level spectra of Ru-Cu₂O and Cu₂O catalyst. Scale bar: a. 2 nm; b. 10 1/nm.

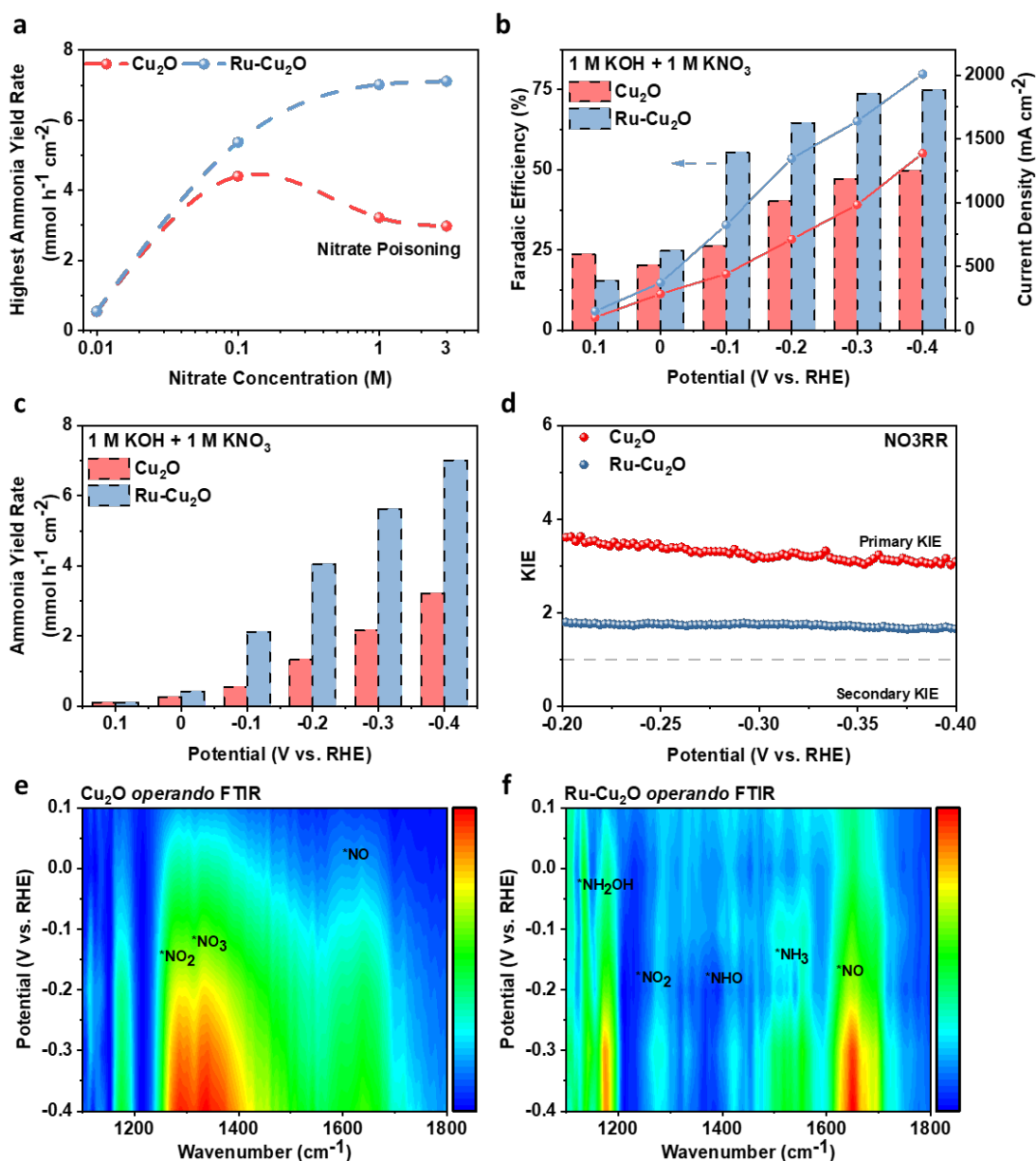


Figure 3. Reaction kinetics of NO_3RR as a function of concentration and voltage in the H-cell.

(a) Optimized ammonia yield rate of various catalysts in an electrolyte containing 1 M KOH and different concentrations of nitrate; (b) Current density and Faradaic efficiency and (c) ammonia yield rate of catalysts in an electrolyte containing 1 M KOH and 1 M KNO_3 ; (d) Comparison in KIE profiles in NO_3RR (in 1 M KOH + 0.1 M KNO_3) using water or deuterium oxide as solvent.; *operando* FTIR measurement for nitrate reduction on the (e) pristine Cu_2O and (f) $\text{Ru-Cu}_2\text{O}$ surface in an electrolyte containing 1 M KOH and 0.01 M KNO_3 .

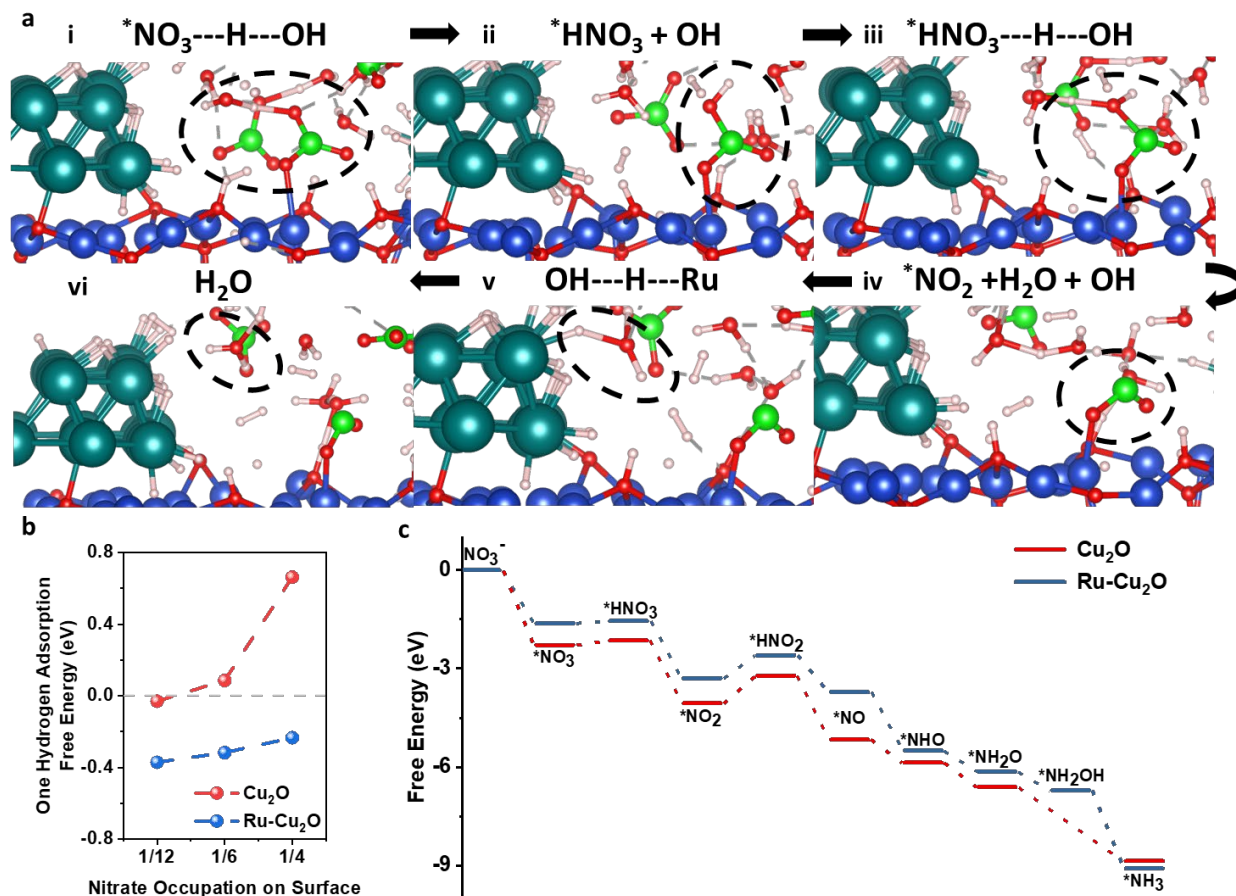


Figure 4. Mechanistic insights on the hydrogen acquisition and transfer. (a) *ab-initio* Molecular dynamic simulation of *NO₃ reduction to *NO₂ and hydrogen transfer process on Ru-Cu₂O catalyst (Time frame: 60, 150, 218, 313, 390, 405 fs). Color scheme: Cu atom in blue, O atom in red, Ru atom in dark green, H atom in white, N atom in light green; (b) Free energy (ΔG) diagram of hydrogen adsorption on Cu₂O and Ru-Cu₂O under various surface coverages of nitrate; (c) Reaction coordinates of NO₃RR on Cu₂O and Ru-Cu₂O catalysts.

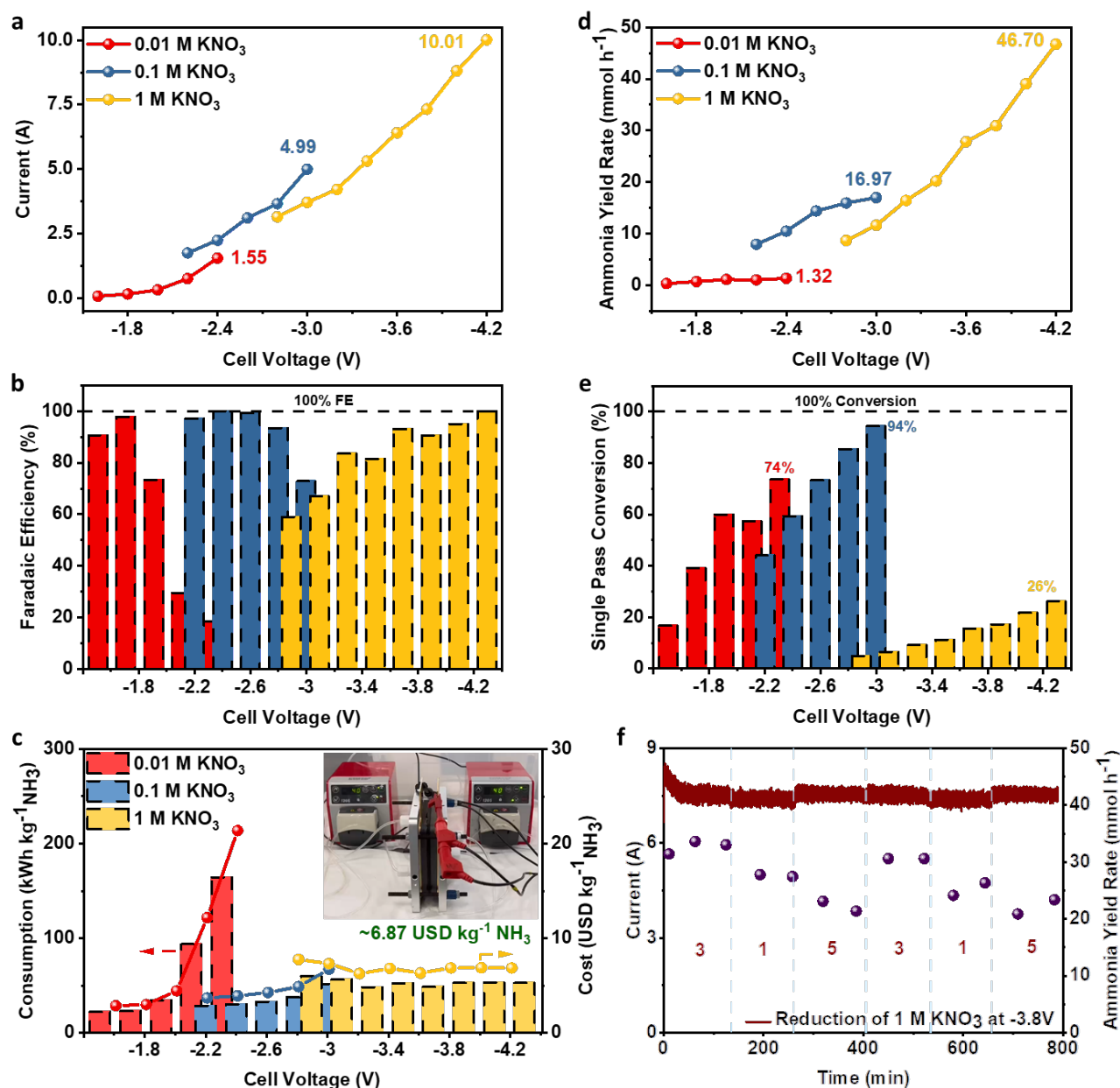


Figure 5. Industrial-level nitrate reduction in a 16 cm^2 flow electrolyzer. Flow cell performance of Ru-Cu₂O catalyst toward nitrate reduction at different concentrations: (a) full cell current; (b) Faradaic efficiency; (c) electricity consumption and production cost. Inset shows the 16 cm^2 flow electrolyzer with two peristaltic pumps; (d) ammonia yield rate; (e) single pass conversion; (f) stability in 1 M KOH and 1 M KNO_3 electrolyte at different flow rates (mL min^{-1}).

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